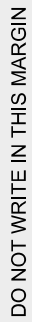


- 5 A doctor is investigating the concentration of blood glucose in patients at risk of developing type 2 diabetes, where blood glucose is measured in appropriate units. The doctor claims that a particular intervention reduces the concentration by more than k units on average. A group of 8 at risk patients is selected at random and each patient follows the intervention for six months. The blood glucose concentrations before and after the intervention are given in the following table.

Patient	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>
Before	183	165	172	165	143	176	161	153
After	164	148	164	149	134	153	155	148

- (a) Use a t -test at the 5% significance level to find the range of values of k for which the result of the test is to reject the null hypothesis. [7]

This image shows a full page of a document template designed for handwritten notes or essays. It features a series of evenly spaced, horizontal grey lines running across the entire width of the page. The lines are thin and light, providing a guide for writing without being distracting. There are no margins, headers, footers, or other markings present on the page.



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This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[1]
